AI & Data Science Projects Report

Project 1: AI-Based Predictive Academic Performance System

Project Overview:

An AI-driven system that predicts students' academic performance and identifies at-risk learners using attendance, assessment scores, engagement levels, and behavioral data.

Skills Required:

- ✓ Python / R Programming
- ✓ Machine Learning (Regression, Classification)
- ✓ Data Preprocessing & Feature Engineering
- ✓ Power BI / Tableau
- ✓ Ethical AI & Fairness Auditing

Target Users:

- ✓ Students
- ✓ Faculty Members
- ✓ Academic Administrators

Expected Outcomes:

- ✓ Early identification of at-risk students
- ✓ Improved academic performance
- ✓ Career pathways in Data Science & AI
- ✓ Ethical and transparent AI implementation

Variables & Data Classification

- ✓ Nominal: Gender, Course Stream
- ✓ Ordinal: Participation Level, Assignment Rating (1–5)
- ✓ Interval: Exam Score (0–100 scale)
- ✓ Ratio: Attendance (%), Study Hours per Week, Number of Assignments Submitted

Data Analysis & Visualization Methods

- ✓ Nominal: Pie charts
- ✓ Ordinal: Stacked bar charts
- ✓ Interval: Line graphs
- ✓ Ratio: Scatter plots

Project 2: AI-Based Smart Disease Risk Prediction System

Project Overview:

A machine learning system that predicts the risk of lifestyle diseases using health metrics and lifestyle data, providing preventive health recommendations.

Skills Required:

- ✓ Machine Learning (Logistic Regression, Decision Trees)
- ✓ Data Cleaning & Preprocessing
- ✓ Statistical Analysis
- ✓ Healthcare Data Ethics Knowledge
- ✓ Dashboard Development

Target Users:

- ✓ Hospitals
- ✓ Fitness Centers
- ✓ Individuals

Expected Outcomes:

- ✓ Early disease risk detection
- ✓ Improved preventive healthcare
- ✓ Career pathways in Healthcare Data Science
- ✓ Ethical handling of medical data

Variables & Data Classification

- ✓ Nominal: Gender, Smoking Status
- ✓ Ordinal: Risk Level (Low/Medium/High), Pain Severity Scale (1–10)
- ✓ Interval: Body Temperature (°C)
- ✓ Ratio: Age, Blood Pressure (mmHg), Glucose Level (mg/dL)

Data Analysis & Visualization Methods

- ✓ Nominal: Bar charts
- ✓ Ordinal: Heatmaps
- ✓ Interval: Trend analysis
- ✓ Ratio: Scatter plots